

GIURGIU, Romania

WMO: 154910

Lat: 43.8753N

Lon: 25.9328E

Elev: 81

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	14.3	6.4	7.6	9.4	11.7	10.0	15.5	31.3	31.1	26.0	37.8	2.6	270	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	94.0	72.5	91.5	71.2	89.0	70.8	76.3	87.2	74.4	85.9	72.9	84.3	4.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	123.1	81.9	71.1	114.9	78.9	69.6	109.0	77.2	39.8	86.2	37.9	86.4	36.6	84.4	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	16.2	12.8	DB	2.3	100.0	4.5	3.6	-0.9	102.6	-3.5	104.7	-6.0	106.8	-9.3	109.4	(4)
(5)			WB	2.0	78.8	4.5	3.6	-1.2	81.4	-3.9	83.5	-6.4	85.5	-9.7	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	30.4	38.3	45.7	55.3	64.6	72.5	75.3	74.9	67.2	54.7	43.2	33.8	(6)	
	DBStd	17.23	8.66	9.16	7.73	7.33	5.94	5.13	4.40	4.82	7.06	7.84	8.58	8.71	(7)	
	HDD50	1938	608	334	176	36	0	0	0	0	42	238	504		(8)	
	HDD65	4817	1073	748	600	297	79	7	0	3	62	327	653	968	(9)	
	CDD50	3671	0	6	41	195	454	675	784	773	517	189	36	1	(10)	
	CDD65	1075	0	0	0	6	68	232	320	311	129	9	0	0	(11)	
	CDH74	11268	0	0	12	155	769	2307	3290	3304	1303	124	4	0	(12)	
	CDH80	4675	0	0	1	30	198	911	1467	1542	499	27	0	0	(13)	
(14) Wind	WSAvg	4.3	4.4	5.0	5.4	5.1	4.5	3.8	3.5	3.4	4.0	3.9	4.1	4.7	(14)	
(15) Precipitation	PrecAvg	24.2	1.7	1.3	1.7	1.8	2.6	2.6	2.4	1.8	2.4	2.0	1.8	2.0	(15)	
	PrecMax	39.1	4.5	4.1	4.1	4.8	6.9	6.6	7.9	5.3	7.8	4.9	4.5	6.5	(16)	
	PrecMin	13.1	0.1	0.0	0.1	0.1	0.6	0.5	0.1	0.1	0.0	0.0	0.0	0.1	(17)	
	PrecStd	6.1	1.0	0.8	0.9	0.9	1.6	1.4	2.0	1.5	2.0	1.3	1.1	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	67.9	75.1	83.8	89.3	95.5	97.2	97.5	93.3	83.6	73.6	58.5	(19)	
		MCWB	47.9	56.2	55.6	64.5	70.6	74.9	74.6	73.0	69.7	66.8	59.5	50.3	(20)	
	2%	DB	49.1	61.3	68.9	78.6	84.9	90.8	93.9	93.8	89.5	77.0	66.4	53.5	(21)	
		MCWB	44.4	52.0	54.3	62.1	66.9	73.0	73.5	71.4	68.5	63.2	57.1	48.1	(22)	
	5%	DB	44.3	56.5	64.6	73.7	82.0	88.2	90.9	91.3	85.7	72.5	60.5	48.6	(23)	
		MCWB	41.4	49.2	52.4	59.9	65.9	72.0	72.3	70.1	67.1	61.6	54.7	45.2	(24)	
	10%	DB	40.7	51.7	59.8	69.7	78.4	85.6	88.0	88.6	82.0	68.2	55.4	44.2	(25)	
		MCWB	39.5	46.4	50.1	57.7	64.6	71.2	71.5	69.9	66.4	59.3	52.2	42.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	56.9	57.7	66.1	72.3	78.2	80.6	77.0	73.5	69.2	60.3	52.6	(27)	
		MCDB	53.7	66.1	71.2	78.7	85.8	89.3	89.0	86.8	87.7	79.5	70.8	57.9	(28)	
	2%	WB	45.0	52.7	55.1	63.6	69.4	75.9	76.9	74.4	71.0	65.1	58.0	48.6	(29)	
		MCDB	49.0	60.2	67.3	75.8	81.0	87.3	87.6	86.8	83.8	73.3	66.0	53.4	(30)	
	5%	WB	41.8	49.8	52.9	61.3	67.3	74.0	74.6	72.9	68.9	62.5	55.2	45.4	(31)	
		MCDB	43.7	55.5	63.3	71.5	78.4	85.2	85.4	85.5	80.8	70.5	59.8	48.3	(32)	
	10%	WB	39.4	46.9	50.7	59.1	65.4	72.4	73.0	71.6	67.3	60.2	52.2	42.2	(33)	
		MCDB	40.8	51.3	59.0	67.7	75.9	82.8	83.7	84.2	78.4	67.6	55.4	44.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	16.8	19.0	21.4	21.7	21.9	22.4	23.8	22.9	19.7	13.2	12.1	(35)	
		MCDBR	17.9	25.9	29.7	30.1	27.3	26.2	27.5	28.7	29.0	27.9	23.0	19.6	(36)	
	5% WB	MCWBR	13.2	17.1	16.4	15.2	11.9	10.8	10.0	9.7	10.9	14.2	13.9	14.4	(37)	
		MCWBR	16.3	25.5	28.1	26.6	24.8	24.1	23.2	24.7	25.6	25.2	21.2	18.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.352	0.385	0.423	0.462	0.437	0.445	0.455	0.469	0.435	0.408	0.379	0.356	(40)	
		taud	2.259	2.185	2.134	2.048	2.199	2.209	2.190	2.148	2.208	2.256	2.288	2.274	(41)	
	Ebn at Noon		236	249	255	255	266	263	259	250	249	238	225	222	(42)	
		Edh at Noon	28	36	43	51	45	44	45	45	40	33	27	25	(43)	
(44) All-Sky Solar Radiation	RadAvg		461	714	1082	1493	1856	2025	2055	1839	1336	867	494	378	(44)	
	RadStd		70	99	106	157	145	144	139	104	104	97	67	53	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-214	+113	N/A	

Nomenclature: See separate page